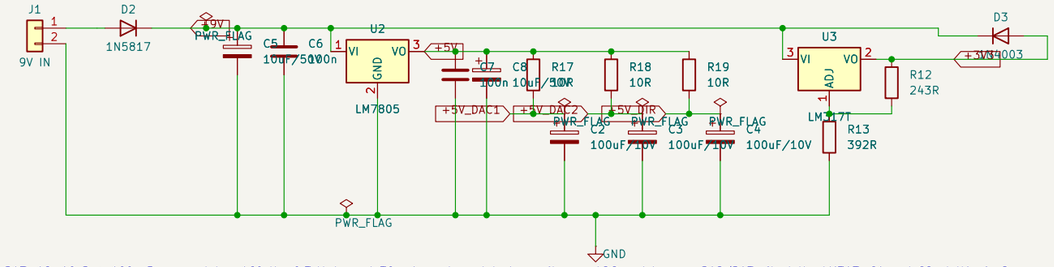
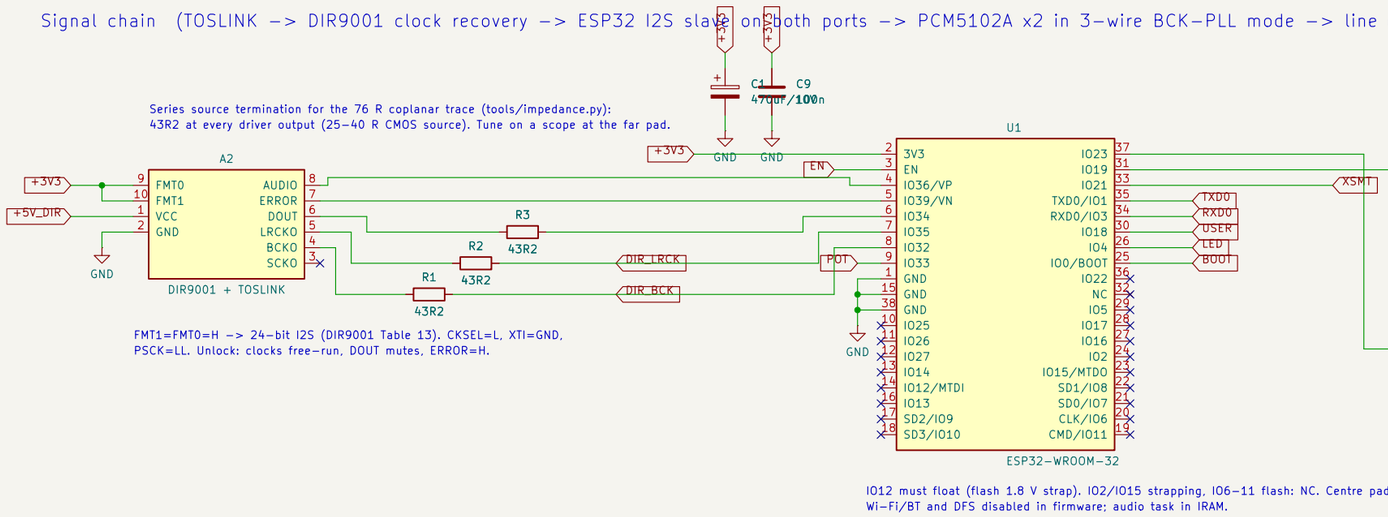
## Power in  (9 V class II adapter -> reverse-polarity Schottky -> LM7805 for the audio modules, LM317 for the ESP32. Star ground at the input.)




R17-19: 10 R + 100 uF per module, ~160 Hz, 0.3 V drop at 30 mA; each module keeps its own LDO and bypass. R12/R13 sit at the LM317; C1 and C9 at U1 pin 2.

## Signal chain  (TOSLINK -> DIR9001 clock recovery -> ESP32 I2S slave on both ports -> PCM5102A x2 in 3-wire BCK-PLL mode -> line out)

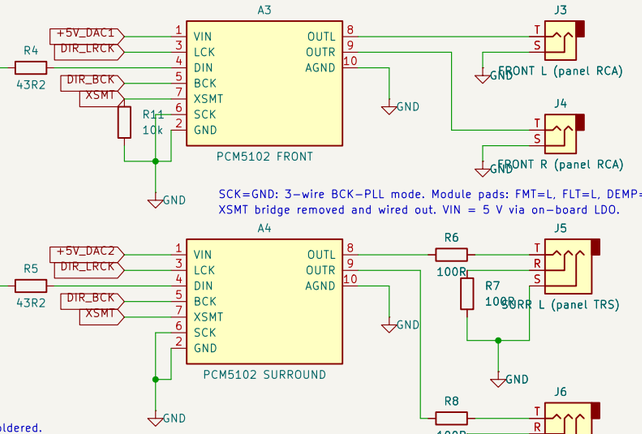



Series source termination for the 76 R coplanar trace (tools/impedance.py):
43R2 at every driver output (25-40 R CMOS source). Tune on a scope at the far pad.

FMT1=FMT0=H -> 24-bit I2S (DIR9001 Table 13). CKSEL=L, XTI=GND,
PSCK=LL. Unlock: clocks free-run, DOUT mutes, ERROR=H.

IO12 must float (flash 1.8 V strap). IO2/IO15 strapping, IO6-11 flash: NC. Centre pad 39 unsoldered.
Wi-Fi/BT and DFS disabled in firmware; audio task in IRAM.

Fronts: panel RCA jacks on screw terminals, short cable to the Fosi amp. Module has 470 R + 2.2 nF out filter.




SCK=GND: 3-wire BCK-PLL mode. Module pads: FMT=L, FLT=L, DEMP=L,
XSMT bridge removed and wired out. VIN = 5 V via on-board LDO.

Panel TRS jacks on screw terminals, impedance-balanced to the Eris 3.5: 100 R tip, 100 R ring to GND.

## ESP32 support and user interface  (no auto-reset: hold BOOT, tap RESET to flash over the UART header. Pot on ADC1, 1k in series because it lives on flying leads.)

